

! Hybrid Cloud and Multi-Cloud Networking

- A hybrid cloud can be used for many business needs:
 - **Application development** – You can develop and test apps in the cloud. After testing, you move the app to the on-premise production system.
 - **Market testing** – You check if people want a new application before spending money on it.
 - **Cloud bursting** – When on-prem resources are full, you can use cloud resources for extra capacity for a short time.
 - **High availability and disaster recovery** – You use both on-prem and cloud environments to keep systems running if one fails.
 - **Compliance** – You can follow rules by keeping sensitive data on-prem and moving less sensitive apps to the cloud.
 - **Archives** – You store old or rarely used data in the cloud at a lower cost.

Reasons for using a multi-cloud infrastructure(altyapı)

- **Features** – You can use special tools or services from different cloud providers.
- **Cost optimization** – You can choose the cheapest service for each workload.
- **Edge locations** – You can run workloads closer to users for better performance.
- **Disaster recovery** – You can keep resources in more than one cloud to protect your system.

• Challenges of hybrid and multi-cloud

- **Service portability** – Not all services work the same in every cloud provider (CSP).
- **Extra connections** – You must create and manage cross-network connections between clouds.
- **More security needs** – More clouds and more connections mean more security measures are required.

Connecting Networks

When you connect your company network to the cloud, the link between your on-premise data center and the cloud is very important for your cloud success. There are **three main ways** to make this connection:

1. **Public Internet** – the simplest method, but not the most secure.
2. **VPN (Virtual Private Network)** – creates a secure, encrypted tunnel between your network and the cloud.
3. **Direct Connect / Interconnect** – a private, high-speed connection that gives better performance and reliability.

These options help organizations build hybrid or multi-cloud networks.

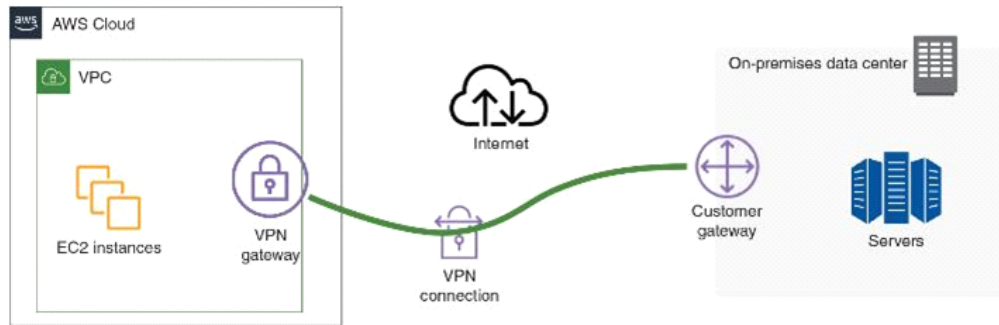


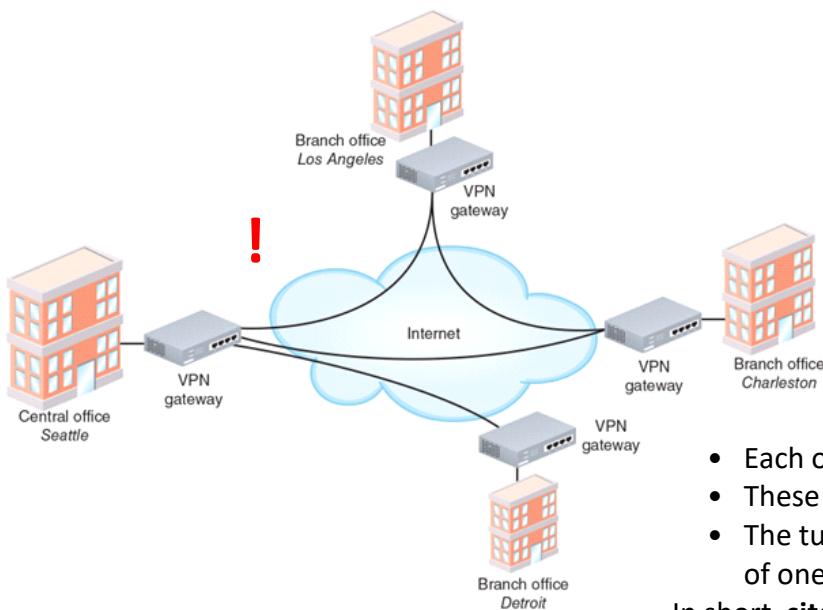
Figure 5-1 A VPN tunnel carries encrypted traffic over the Internet between two VPN gateways

Virtual Private Network

- A **VPN (Virtual Private Network)** creates a secure, encrypted tunnel so data can travel safely over the public Internet.
- To use a cloud VPN, your **VPC must have VPN endpoints**. Users connect by using a client configuration file that you provide.
- If you want to connect **one VPC to another**, each VPC needs a **VPN gateway (VPN endpoint)**.
- For better reliability and higher availability, you can create **multiple VPN tunnels** on each endpoint.

VPN Tunneling Protocols

- VPN tunneling protocols wrap higher-level data inside another protocol. This process is called **tunneling**, and it creates a virtual link between two VPN endpoints.
- Some tunneling protocols work at **Layer 2** (data link layer), while others work at **Layer 3** (network layer).
- Most VPNs use **Ipsec(Internet Protocol Security)** to secure data and keep it private.
- Common VPN tunneling protocols include:
- **PPTP, L2TP, GRE, OpenVPN, IKEv2, and WireGuard.**



- Each office has a **VPN gateway device**.
- These gateways create secure **VPN tunnels** between the sites.
- The tunnels allow all offices to communicate as if they are part of one private network.

In short, **site-to-site VPN** connects many branch offices together securely over the Internet.

Figure 5-4 A site-to-site VPN gateway connects each site to one or more other sites

This diagram shows a **site-to-site VPN topology**.

In this setup, different office locations (Seattle, Los Angeles, Detroit, Charleston) are **connected to each other** through the **Internet** using **VPN gateways**.



Figure 5-5 Point-to-site remote clients connect to the network through the VPN headend device

This diagram shows a **point-to-site VPN**.

Remote users—like a mobile user, a home worker, or a small remote office—connect to the company network through the **VPN headend** over the Internet.

Each user connects individually, instead of connecting whole offices together.

Direct Connect / Interconnect

- A **direct connection** links your data center to the cloud through a shared location called a **colocation (colo)**.
- A colo is a data center where many service providers connect with their customers.
- By using the colo, you can create fast, private connections to many cloud or service providers, which helps build **hybrid** and **multi-cloud** systems.

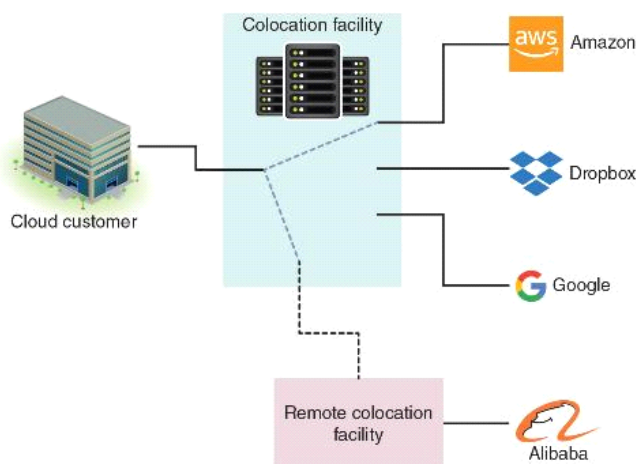


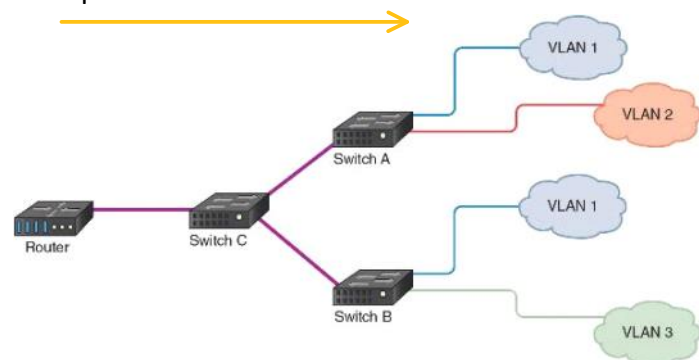
Figure 5-6 A colocation facility for creating direct connections between service providers and their customers

! Cloud Peering

- **Cloud peering** connects multiple VPCs so they can talk using **private IPs**, not the public Internet.
- It gives faster and more secure communication between VPCs in the same company or even between different companies.
- Many cloud providers offer a central device (like AWS Transit Gateway) to easily connect many VPCs together

Virtual LAN (VLAN)

- A **VLAN groups** switch ports so devices in different VLANs must use a router to communicate.
- This helps the admin control which devices can talk to each other without using extra switches.
- A **VLAN tag** shows which VLAN a message belongs to and sends it only to that VLAN.
- VLANs are also used to manage direct links between on-prem data centers and cloud providers.



seperates the traffic
increase the security
blocks unnecessery communication
makes network management easier

Figure 5-11 Three switches on a LAN with multiple VLANs

DHCP Services

- Dynamic IP addresses are assigned by a **DHCP server**.
- With the IP address, the device also receives extra information:
 - A time limit called a **lease time**
 - The IP address of the router (the **default gateway**)
 - **Primary and secondary DNS** server addresses to resolve domain names
- A network admin can also set **reserved** or **excluded** IP addresses.

DHCP in the Cloud

- When you create a VPC, you choose an address space. **New VMs get their IP addresses from this space** (the IP address range (CIDR block) *you set* when creating the VPC.).
- The cloud automatically gives IP addresses in the background.
- After setting the CIDR block, you don't need to do anything else for VMs to get private IP addresses.
- You can change some DHCP options that a VM receives when its vNIC is created.

- **You choose the IP range** when you set the **CIDR block** of the VPC.
- This CIDR block becomes the **pool of IP addresses** for the VPC.
- **DHCP automatically assigns an IP address** from this pool to each new VM in the background. (detailed expl. for myself)

DHCP Across a Hybrid Cloud

- **IPAM**(IP Address Management) is a tool that can manage DHCP for both on-prem and cloud networks from one system
- IPAM helps plan, assign, and track all IP addresses in the network
- Public clouds like AWS and Azure do not include full IPAM features
- Third-party IPAM tools can work with cloud platforms and on-prem networks in hybrid and multi-cloud setups

DNS Services

- **DNS matches computer names to IP addresses**

It converts names (like google.com) into IP addresses so computers know where to send traffic.

- **DNS servers store lists of names and their IP addresses**

These records are kept in large distributed databases around the world.

- **DNS in the Cloud**

– All AWS VPCs use Amazon's DNS server by default

Every instance automatically receives AWS's DNS unless changed.

– You can configure your VPC to use your own DNS or public DNS

By creating a new DHCP options set, you can point instances to any DNS server you want.

– Route 53 allows on-prem networks to use cloud DNS

Your physical office network can reach AWS DNS through a VPN or Direct Connect link.

Routing

- **Routing protocols are divided into two main groups**

– Interior Gateway Protocols (IGPs): handle routing *within* one autonomous system (AS)

– Exterior Gateway Protocols (EGPs): handle routing *between* different autonomous systems

- **Common Interior Gateway Protocols (IGPs)**

- RIP (Routing Information Protocol) and RIPv2
- OSPF (Open Shortest Path First)
- IS-IS (Intermediate System to Intermediate System)
- IGRP (Interior Gateway Routing Protocol)
- EIGRP (Enhanced Interior Gateway Routing Protocol)

VPN Routing

– In a VPN connection (whether on-prem ↔ cloud or cloud ↔ cloud), In the cloud, you usually must use BGP, an **exterior** gateway protocol

– BGP dynamically manages routes between autonomous systems (AS)

– To use BGP, the device on both ends of the VPN must support it

Direct Connect Routing

A **Direct Connect** is a *physical, private* connection from your on-prem network to the cloud (not over the public Internet like VPN).

– Direct Connection services requires BGP routing on each virtual interface

– One Direct Connect link can have many virtual interfaces, each with its own VLAN

!! Load Balancing

- **Load balancing** distributes traffic across multiple servers to improve performance and prevent failures.
- **How it works**
 - The load balancer sits between the Internet and the web server cluster.
 - It receives incoming traffic and forwards it to the servers.
- **Load-balancing algorithms**
 - The load balancer selects a server using a specific algorithm.
 - **Example:** Round-robin → each server gets traffic in a repeating order.
 - More advanced algorithms can be configured through a **PAC file (Proxy Auto-Config)**.

Load Balancing in the Cloud

- Cloud providers offer different load balancers, usually based on the OSI layer they operate in.
- For example, Azure currently offers only a **Layer 4** load balancer.

Load Balancing Across a Hybrid Cloud

- Cloud load balancers can extend to support servers running **on-premises** or in **other clouds**.
- **GSLB (Global Server Load Balancing)** can route users to the closest server for better performance.